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A Simulation Framework for Evaluating Spatial Regression Models of Hierarchical Areal Data

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Abstract. This study constructs a controlled simulation environment to rigorously examine the behaviour of the CAR model under diverse spatial conditions. Using an artificial study region designed to emulate real-world areal heterogeneity, the analysis evaluates how CAR responds to varying spatial data-generating processes, specifically those governed by CAR, SAR+ CAR, and GWR frameworks. The experiments focus on how well CAR captures underlying spatial dependence, how its estimates behave under model misspecification, and how its predictions compare across settings that differ in spatial structure. By systematically varying these conditions, the simulation results illuminate the reliability, strengths, and potential limits of CAR-based inference, laying the groundwork for its later application to real-world housing data.

Key words and Phrases: Conditionally autoregressive, hierarchical data, simulation study, spatial dependence.

1. INTRODUCTION

Spatial data are increasingly prevalent in many applied fields, including epidemiology, environmental science, and real estate analysis [1], where observations are often geographically referenced and exhibit dependence across space. In such settings, ignoring spatial autocorrelation can lead to biased inference and misleading predictions. Consequently, a wide range of spatial regression models has been developed to account for different forms of spatial dependence [2, 3, 4].

Among these, the Conditionally Autoregressive (CAR) model first introduced by [5] has become one of the most widely used approaches for areal data, where spatial units are defined by discrete regions such as districts or administrative

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boundaries [6, 7, 8, 9]. The CAR framework captures spatial dependence through neighbourhood structures, allowing adjacent areas to share similar random effects. This makes it particularly suitable for modelling phenomena driven by local interactions and spatial clustering [10, 11].

However, spatial processes encountered in practice are often more complex than purely areal dependence. In many applications, spatial variation may arise from multiple mechanisms simultaneously, including lag-based dependence, as represented by Spatial Autoregressive (SAR) models [12, 13], and smoothly varying relationships, as captured by Geographically Weighted Regression (GWR) [14, 15, 16, 17]. These alternative modelling approaches reflect different underlying assumptions about how spatial dependence operates, and their performance may vary substantially depending on the true data-generating process.

Despite the widespread use of these models, there remains limited understanding of how the CAR model behaves under model misspecification, particularly when the true spatial process deviates from a purely areal structure. In addition, the relative performance of CAR compared to other spatial and non-spatial models across different spatial scenarios has not been fully explored in a controlled simulation environment.

This study addresses these gaps by constructing a comprehensive simulation framework designed to evaluate the behaviour of spatial regression models under varying spatial data-generating processes. An artificial study region is developed to mimic realistic spatial heterogeneity, within which both point-level and area-level datasets are generated. The response variable is simulated under three distinct mechanisms: a CAR process representing areal dependence, a hybrid SAR+CAR process capturing both lag and local effects, and a GWR process representing smoothly varying spatial relationships. This work extends an earlier study [18] in which a preliminary version of the simulation was used for model justification, by developing a more general framework for systematic comparison across spatial processes and data structures.

Using this framework, we systematically compare the performance of several models, including CAR, multilevel CAR (mlvCAR), SAR, GWR, generalized linear models (GLM), and generalized linear mixed models (GLMM). The evaluation focuses on parameter recovery, uncertainty, spatial dependence estimation, and predictive accuracy across multiple simulation replications.

The contributions of this study are threefold. First, it provides a controlled and flexible simulation environment for analysing spatial models under diverse dependence structures. Second, it offers a systematic assessment of the robustness of CAR-based models under both correct specification and misspecification. Third, it highlights the conditions under which different modelling approaches are most appropriate, thereby offering practical guidance for applied spatial analysis.

The remainder of this paper is organised as follows. Section 2 describes the simulation design, including the construction of the artificial spatial region, covariate generation, and data-generating processes. Section 3 presents the simulation

results and comparative analysis. Finally, Section 4 concludes with key findings and implications for future research.

2. SIMULATION DESIGN

Simulation studies are widely used in statistical methodology to evaluate model performance under controlled data-generating mechanisms, particularly when analytical comparisons are intractable [19, 20, 21]. We construct an artificial study region partitioned into multiple areal units. Each area contains several observation points with known planar coordinates (x, y), and at every point we record four covariates X_1, X_2, X_3, and X_4.

Stacking all points yields a dataset with N rows (one row per point), and columns for area/point identifiers, coordinates, and the covariates. In other words, the design has $k = 4$ covariates $(X_1, \ldots, X_4)$ in addition to the coordinates and IDs.

This point-level dataset is then used to generate the response Y under three alternative data-generating scenarios/process (DGP); the specific mechanisms and parameter settings are described later.

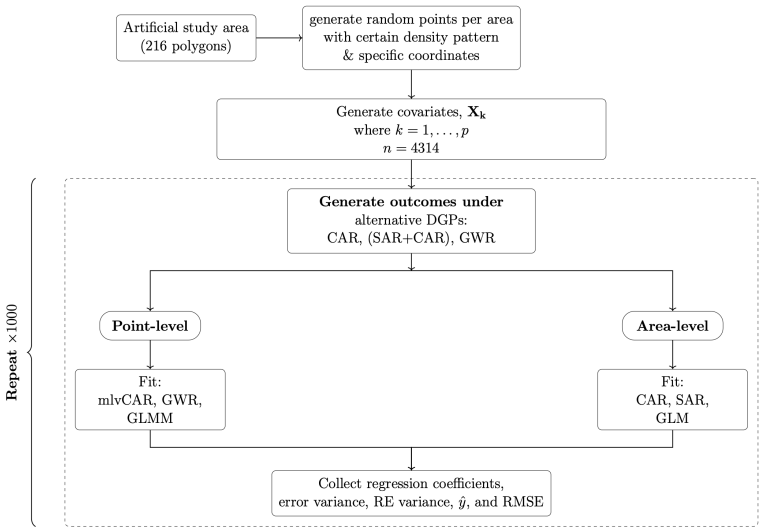

FIGURE 1. Proposed simulation framework. Spatial data are generated under multiple mechanisms (GWR, CAR, and SAR+CAR) and represented at both areal and point levels. A range of competing models is fitted to each dataset, and their performance is systematically evaluated across repeated simulations using bias and RMSE.

2.1. **Artificial Study Region.** The experiment begins with the construction of an artificial spatial region consisting of 216 polygons, each representing a distinct areal unit. We draw $K = 216$ seed points uniformly inside a Lombok bounding window (WGS 84 [1]; [116.0°, 116.5°]E, [8.5°, 8.0°]S) and apply a **Voronoi**[2] tessellation to obtain an irregular lattice [22]. The tessellation is then clipped to the window, small sliver cells are removed using a minimum–area rule, and the result is stored as an `sf` layer with CRS EPSG:4326. [3] This yields an areal partition of 216 polygons with realistic variation in size and number of neighbours.

A map of the final lattice and its neighbourhood graph is shown in Figure 2 and serves as the fixed spatial support for all simulations.

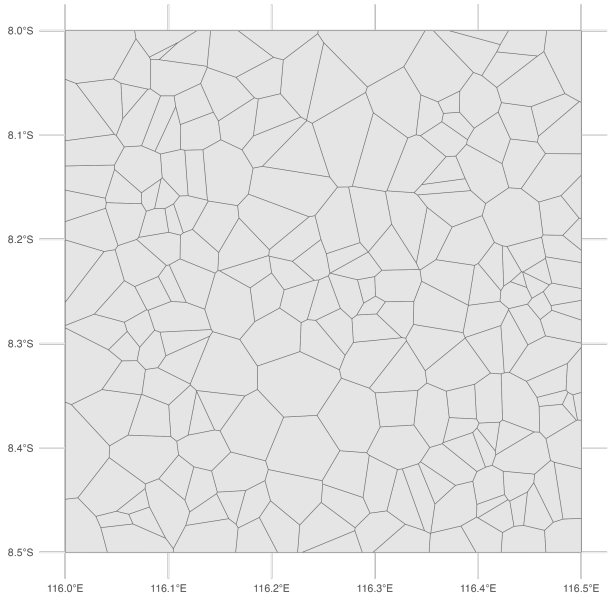

FIGURE 2. Artificial areal partition (216 units). Voronoi polygons clipped to a bounding box; grey lines show polygon boundaries.

The resulting neighbour graph for the artificial region is shown in Figure 3. Edges connect polygons that are neighbours, providing a visual summary of the spatial connectivity that underlies the simulations.

[1]World Geodetic System, a global standard for measuring locations on Earth.
[2]A spatial partition where each region consists of all points closer to one seed than to any other
[3]Coordinate reference system - European petroleum survey group. It is a coordinate reference system defined by an EPSG code, such as EPSG:4326 for WGS 84.

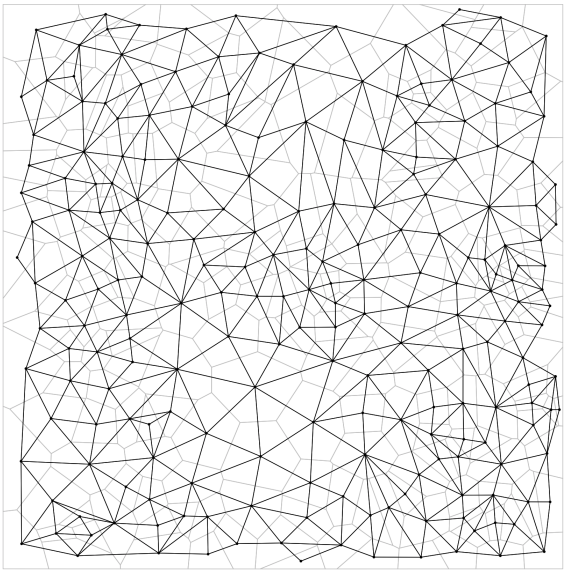

FIGURE 3. Neighbourhood structure of the artificial region. This figure illustrates the spatial connectivity between polygons in the artificial study region, where each line represents a defined neighbour relationship based on spatial adjacency

2.2. **Synthetic density and point generation.** To emulate realistic spatial variation, the artificial region is endowed with a density pattern in which a central horizontal belt exhibits higher housing density, an abstraction inspired by Lombok's population and development gradient. Density is not used as a covariate in subsequent models; instead, it controls how many observation points each polygon receives.

Let y_i be the y-coordinate of the centroid of area i. We define a smooth ridge centered at $\bar{y}$ with Gaussian decay and mild noise,

$$d_i^\star \;=\; \exp\!\left\{-\frac{(y_i-\bar{y})^2}{2\sigma^2}\right\} \;+\; \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0,\sigma_\varepsilon^2),$$

truncate below a small floor, and rescale to $[500, 1000]$:

$$d_i \;=\; 500 \;+\; 500\,\frac{d_i^\star - \min(d^\star)}{\max(d^\star)-\min(d^\star)} \;\in\; [500, 1000].$$

For each area i, the number of point locations is a bounded affine function of d_i,

$$n_i \;=\; \mathrm{clip}\!\left\{\; \left\lfloor m_{\min} + s\,(m_{\max}-m_{\min})\,\frac{d_i-500}{1000}\right\rceil,\; m_{\min},\; m_{\max}\right\},$$

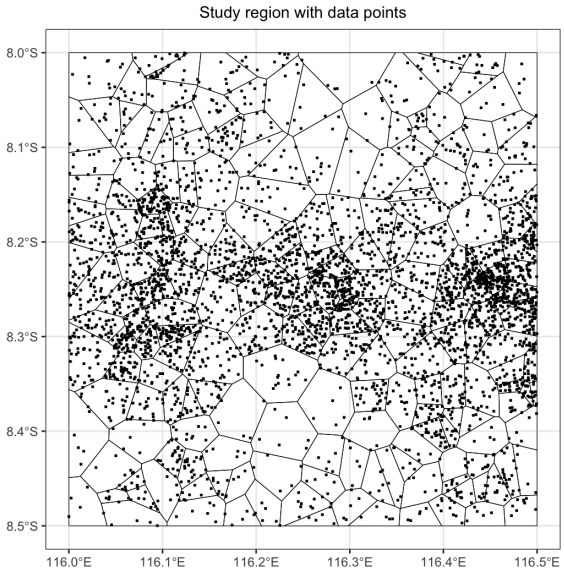

FIGURE 4. Artificial study region with observation points. The points represent data locations within each polygon, simulating real-world spatial observations for the purpose of model analysis and validation.

with $m_{\min} = 5$, $m_{\max} = 150$, and scaling factor $s = 0.7$. Points are then sampled uniformly within each polygon. In practice, this yields 4,314 points across **216** polygons, with **5–56** points per polygon, reflecting higher concentration in the central belt. Figure 4 shows the resulting point-level distribution.

In practice, polygons with higher density yield more observations, mimicking the fact that densely populated urban centres typically host more housing transactions. A total of 4,314 points were generated across the 216 polygons, with each polygon containing between 5 and 56 observations. Figure 4 depicts this point-level distribution.

2.3. **Covariate Generation.** Each observation point is assigned four structural attributes used as explanatory variables for prices: land size (X_1), building size (X_2), bedrooms (X_3), and bathrooms (X_4). To induce realistic co-movement, the point-level vector $\mathbf{X}_i = (X_{i1}, X_{i2}, X_{i3}, X_{i4})^\top$ is drawn from a multivariate normal distribution:

$$\mathbf{X}_i \sim \mathcal{N}_4(\,\boldsymbol{\mu},\ \boldsymbol{\Sigma}\,), \qquad \boldsymbol{\mu} = (300,\ 120,\ 3,\ 2)^\top.$$

The covariance is built from standard deviations $\{50,\ 30,\ 1,\ 1\}$ and a target correlation matrix $\mathbf{R}$ (e.g., $\rho_{12} = 0.8$, $\rho_{13} = 0.5$, $\rho_{23} = 0.6$, $\rho_{34} = 0.7$):

$$\mathbf{\Sigma} = \mathbf{D}\,\mathbf{R}\,\mathbf{D}, \qquad \mathbf{D} = \mathrm{diag}(50,\ 30,\ 1,\ 1).$$

Draws are mapped to plausible ranges: $X_1 \in [100, 1500]$ m^2, $X_2 \in [40, 800]$ m^2, $X_3 \in \{1, \ldots, 5\}$, and $X_4 \in \{1, \ldots, 3\}$ (by rounding and truncation). A representative slice is shown in Table Table 1.

For areal analyses, point-level covariates are averaged within polygons:

$$\bar{X}_{Aj} \;=\; \frac{1}{n_A} \sum_{i \in A} X_{ij}, \qquad j = 1, \ldots, 4,$$

where A indexes polygons and n_A is the number of points in polygon A. As a diagnostic, Moran's I (1000 permutations) is computed on these area-level means to gauge residual spatial autocorrelation introduced by aggregation.

A representative slice of this dataset is shown in Table 1.

Table 1: Slice of generated dataset as covariates for simulations

ID	Land size	Building size	Bedrooms	Bathrooms
1	270.48	108.55	4	2
1	295.99	100.13	4	2
2	319.52	126.31	3	1
2	325.01	120.84	4	3
2	294.15	144.05	4	3
3	222.18	81.4	3	2
4	414.23	150.15	2	2
4	278.75	101.41	2	1
⋮	⋮	⋮	⋮	⋮

2.4. **Generating Y: Three different DGPs.** Given the covariates $\mathbf{X}$ and the artificial areal lattice, we simulate the response Y under three spatial DGPs: CAR, SAR+CAR, and GWR. All three data-generating processes use the same covariate standardisation, neighbourhood support, and output construction.

For point j in area a and covariate $k \in \{1, 2, 3, 4\}$,

$$X^{\mathrm{std}}_{ajk} \;=\; \frac{X_{ajk} - \bar{X}_k}{s_k},$$

where $\bar{X}_k$ and s_k are the global mean and standard deviation computed over all points. The point-level design vector includes an intercept,

$$\mathbf{x}_{aj} \;=\; \left(1,\ X^{\mathrm{std}}_{aj1},\ X^{\mathrm{std}}_{aj2},\ X^{\mathrm{std}}_{aj3},\ X^{\mathrm{std}}_{aj4}\right)^{\top}.$$

The artificial lattice and polygon contiguity are fixed across scenarios (Figure 3). The binary adjacency matrix is $\mathbf{W}$ with diagonal matrix $\mathbf{D} = \mathrm{diag}(d_{aa})$, $d_{aa} = \sum_b w_{ab}$.

Coefficient values are calibrated from a beta function that is inspired from GWR concept about spatially varying coefficients as a function of location coordinate, where in this context it refer to areas centroid (x_a, y_a), with components specified as trigonometric–linear trends plus small Gaussian noise (Equation 1)

$$\beta_k(\mathbf{s}_a) \;=\; b_k \;+\; c_{k1}\,\sin(dx_a) \;+\; c_{k2}\,\cos(dy_a) \;+\; c_{k3}\,x_a \;+\; c_{k4}\,y_a \;+\; \varepsilon_k, \qquad (1)$$

The *sine–cosine* terms create smooth, oscillating spatial patterns, the linear terms impose gentle large-scale gradients, and the small Gaussian perturbations ε_k add local variability while keeping the surfaces smooth (smaller variance for β_3, β_4 yields tighter fields for room counts). Evaluating these functions at polygon centroids yields area-specific coefficients that vary continuously across the map but remain constant within each polygon, which is consistent with the GWR design used in the simulations. values in Equation 2 are the mean column of beta matrix. it yields:

$$\boldsymbol{\beta} = (1,\ 0.9,\ 0.7,\ 0.5,\ 0.3)^\top, \qquad \sigma_\varepsilon^2 = 0.2, \qquad \sigma_\phi^2 = 0.6, \qquad \rho = 0.6. \qquad (2)$$

The idiosyncratic noise $\sigma_\varepsilon^2 = 0.2$ and the spatial variance $\sigma_\phi^2 = 0.6$ imply a spatial-to-nugget variance ratio of approximately 3:1. This indicates that spatial structure is clearly present, yet does not dominate the mean signal. The spatial dependence parameter $\rho = 0.6$ induces moderate smoothing—strong enough to capture local spillover effects while avoiding numerical instability near the unit boundary. Using the same parameter set $(\boldsymbol{\beta}, \sigma_\varepsilon^2, \sigma_\phi^2, \rho)$ for CAR, SAR+CAR, and GWR ensures that any observed differences are attributable to the dependence structure rather than differences in signal strength.

Each scenario constructs a log-scale outcome ℓ_{aj}, which is then transformed back to the original scale via

$$Y_{aj} = \exp(\ell_{aj}).$$

From each simulated dataset, we export:

$$\text{Point level:} \quad \{Y_{aj}, \ell_{aj}, \mathbf{x}_{aj}, a\},$$

$$\text{Area level:} \quad \left\{ \bar{X}_{Ak} = \frac{1}{n_A} \sum_{j \in A} X_{Ajk},\ \tilde{Y}_A = \text{median}\{Y_{Aj}\} \right\}.$$

For comparability, area-level covariates are re-standardised using the same global $(\bar{X}_k, s_k)$.

Scenario 1: GWR DGP (Spatially Varying Coefficients)

In this scenario, data are generated under the GWR framework, which allows relationships between the response variable and explanatory variables to vary across space. Unlike global regression models that assume a constant parameter vector $\boldsymbol{\beta}$, the GWR model assigns location-specific coefficients $\boldsymbol{\beta}(\mathbf{s}_i)$ to each spatial point i, enabling local variation in the relationships.

The coefficient surfaces $\boldsymbol{\beta}(\mathbf{s}_i)$ vary smoothly over space as functions of spatial coordinates (x_i, y_i), with the addition of small random perturbations to introduce spatial heterogeneity. The structure follows Equation 1.

Using these locally varying coefficients, the response variable is generated as

$$Y_i = \beta_0(\mathbf{s}_i) + \beta_1(\mathbf{s}_i)X_{i1} + \beta_2(\mathbf{s}_i)X_{i2} + \beta_3(\mathbf{s}_i)X_{i3} + \beta_4(\mathbf{s}_i)X_{i4} + \varepsilon_i, \tag{3}$$

which is derived from Equation **??**, where $\varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ represents independent random noise. This formulation ensures that each observation reflects a locally adaptive relationship, producing a spatially heterogeneous dataset suitable for evaluating the ability of GWR to recover spatially varying coefficients.

Scenario 2: CAR DGP (Areal Random Effects)

Hete, the spatial effect $\boldsymbol{\phi}$ plays a central role in inducing spatial dependence among areas. The neighbourhood structure is defined by the spatial adjacency matrix $\mathbf{W}$, where two areas are considered neighbours if they share a common boundary.

The spatial random effects $\boldsymbol{\phi}$ are generated from a multivariate Gaussian distribution following the Leroux prior:

$$\boldsymbol{\phi} \mid \mathbf{W}, \tau^2, \rho, \boldsymbol{\mu} \sim \mathcal{N}\Big(\boldsymbol{\mu},\ \tau^2\left[\rho\mathbf{W}^* + (1-\rho)\mathbf{I}_n\right]^{-1}\Big), \tag{4}$$

where $\mathbf{W}^* = \mathbf{D} - \mathbf{W}$ and $\mathbf{D}$ is a diagonal matrix containing the number of neighbours for each area. The matrix $\mathbf{Q} = \rho\mathbf{W}^* + (1-\rho)\mathbf{I}_n$ acts as a precision matrix controlling spatial smoothness. Smaller values of ρ imply weaker spatial dependence, whereas values close to 1 produce stronger spatial autocorrelation.

Once $\boldsymbol{\phi}$ is generated, it is incorporated into the response according to Equation **??**, yielding spatially correlated outcomes that reflect the underlying spatial structure.

Scenario 3: SAR + CAR DGP (Lagged Mean and Areal Effects)

This scenario combines both SAR and CAR structures in a two-stage process.

In the first stage, the response vector $\mathbf{Y}$ is generated from its joint distribution, as described in Equation **??**. This approach is computationally efficient compared to directly simulating from the recursive SAR formulation (Equation **??**), where each observation depends on neighbouring values. The covariance matrix is given by

$$\sigma^2(\mathbf{I} - \rho\mathbf{W})^{-1}(\mathbf{I} - \rho\mathbf{W})^{-T},$$

where ρ is the spatial autoregressive parameter, $\mathbf{I}$ is the identity matrix, and $\mathbf{W}$ is the spatial weight matrix.

In the second stage, spatial random effects $\boldsymbol{\phi}$—as defined in Equation 4—are added to the SAR-generated outcomes:

$$\mathbf{Y}_{\text{SAR+CAR}} = \mathbf{Y}_{\text{SAR}} + \boldsymbol{\phi}.$$

This hybrid construction captures both global spatial dependence (via the SAR component) and localised spatial heterogeneity (via the CAR component), producing data that more closely resemble complex spatial processes observed in practice.

2.5. **The Simulations.** After all simulation components were prepared, the experiments were conducted in two iterative sets of $M = 1000$ replications. Each set corresponds to a different level of analysis: point-level and area-level. The first set (point-level data) involved fitting the mlvCAR, GLMM, and GWR models. The second set (area-level data) involved fitting the CAR, SAR, and GLM.

For the GWR model, the fitting procedure requires spatial coordinates for each observation. Therefore, the coordinates used were the artificial ones generated during the observation point generation stage (see Section 2.2). This approach reflects practical constraints in real-world settings, where it is often difficult to obtain housing transaction data from web listings that include complete addresses and precise geographic coordinates.

The CAR and mlvCAR models were estimated using the `CARBayes` package, specifically via the functions `S.CARleroux` and `S.CARmultilevel`, which implement Bayesian inference under the Leroux specification. For the non-spatial counterparts, the GLMM and GLM models were fitted using the `rstanarm` package (`stan_glmer`) and the base R function `glm`, respectively. This allows for a direct comparison between spatial and conventional hierarchical modelling approaches. Spatial dependence based on lag structures and spatially varying relationships was captured using the SAR and GWR models, implemented via `lagsarlm` and `gwr`, respectively.

From each simulation, the extracted quantities include the estimated means of the intercept, regression coefficients, and spatial parameters, such as $\hat{\rho}$, $\hat{\sigma}^2_\varepsilon$, and $\hat{\sigma}^2_\phi$. In addition, the corresponding standard deviations and confidence intervals were computed. The root mean squared error (RMSE) was also calculated for each iteration as a complementary measure of model performance.

The simulation outputs were stored in a structured list object using the `furrr` package to enable parallel computation across iterations. Each simulation run produced a set of outputs, including estimated parameters, spatial parameters, and prediction errors. These outputs were combined using the `do.call()` and `rbind()` functions to form unified data frames for subsequent analysis.

Specifically, the main simulation object `sim_results` contains results from all $M = 1000$ iterations. Each element of this object includes:

- `params`: estimated regression coefficients and intercepts,
- `params_sp`: spatial parameters such as $\hat{\rho}$, $\hat{\sigma}^2_\varepsilon$, and $\hat{\sigma}^2_\phi$, and
- `rmses`: RMSE values for model performance evaluation.

These components were subsequently extracted and consolidated into three separate data frames, namely `params_df`, `paramssp_df`, and `rmse_df`, representing the regression estimates, spatial parameters, and predictive performance metrics, respectively.

3. Results

This section presents the outcomes of the simulation experiments, focusing on the comparative performance of the considered models under different spatial

data-generating processes. The analysis evaluates both estimation accuracy and predictive capability across point-level and area-level datasets. In particular, attention is given to the behaviour of regression coefficients, the magnitude of bias, and overall model fit as measured by predictive error. The results are organised into two main components: (i) bias and predictive performance, and (ii) the behaviour of the spatial dependence parameter ρ across scenarios.

3.1. **Bias and Predictive Performance.** Figures 5 and 6 present the bias estimates of regression coefficients across different modelling approaches.

Under the CAR data-generating process (DGP), all three models exhibit relatively low bias, indicating that each is capable of recovering the true parameter values with reasonable accuracy. However, their levels of uncertainty differ substantially. The GWR model displays the widest confidence intervals, reflecting its limited ability to capture the discrete areal dependence inherent in the CAR process. In contrast, both the GLMM and mlvCAR models produce tighter confidence bands and more stable estimates. In particular, the mlvCAR model aligns closely with the data-generating mechanism, as it explicitly incorporates hierarchical areal effects similar to those in the CAR structure. This suggests that models designed to represent spatial random effects yield more efficient and reliable estimates when the underlying process follows an areal random-effect structure.

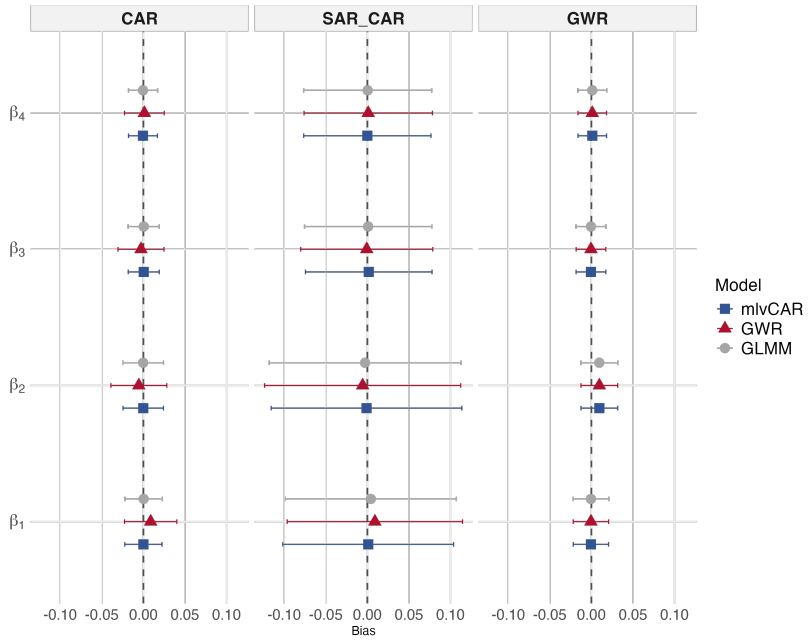

FIGURE 5. Bias values by method and dataset type for point-level data

Under the SAR+CAR DGP, which combines spatial lag dependence with areal random effects, all models exhibit increased variability and bias compared to the previous scenario, reflecting the higher complexity of the underlying spatial process. The mlvCAR model tends to produce the most stable estimates, although with wider uncertainty bands, indicating that it captures areal correlation effectively but does not fully represent the lag dependence component. The GWR and GLMM models show greater bias and wider intervals, particularly for coefficients related to building and land size, highlighting their limitations in modelling joint spatial lag and random-effect structures. This scenario underscores the challenges associated with hybrid spatial dependencies and the need for models that explicitly integrate both components.

When data are generated under the GWR DGP, where spatial variation is expressed through smoothly varying local relationships, the GWR model performs as expected, yielding the smallest bias and narrowest confidence intervals across all coefficients. This confirms its effectiveness in capturing continuous spatial heterogeneity. The GLMM also produces relatively unbiased estimates, albeit with slightly larger dispersion, as it does not explicitly account for spatial dependence. In contrast, the mlvCAR model remains stable and largely unbiased but shows broader uncertainty than GWR, reflecting its approximation of continuous variation using discrete areal random effects. Nevertheless, mlvCAR performs robustly, indicating that hierarchical spatial structures can partially capture smooth spatial variation.

For the area-level datasets, bias patterns remain generally small across all models, suggesting that each approach is capable of recovering the true parameters with reasonable accuracy. However, distinct differences emerge depending on the underlying DGP.

Under the CAR DGP, the CAR, GLM, and SAR models all produce nearly unbiased estimates, with values centred around zero and relatively narrow confidence intervals. The CAR model performs best in this setting, as expected, due to its direct alignment with the underlying areal dependence structure. The GLM and SAR models also provide stable estimates, although the SAR model exhibits slightly greater variability, likely due to its lag-based formulation.

Under the GWR DGP, bias remains close to zero across all models, indicating robustness even when the true process follows continuous spatial variation. However, confidence intervals are slightly wider, particularly for the SAR model, reflecting structural mismatch between the assumed and true processes.

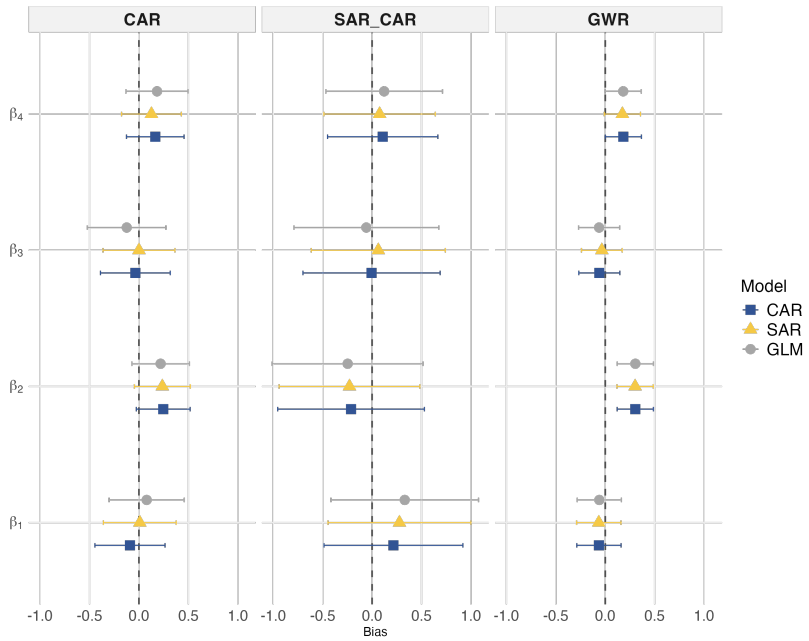

FIGURE 6. Bias values by method and dataset type for area-level data

In contrast, the SAR+CAR DGP produces the largest variation among models. The SAR model remains relatively unbiased but shows higher uncertainty, benefiting from its ability to capture lag dependence. The CAR model performs consistently, although its confidence intervals widen slightly due to the added complexity. The GLM, lacking an explicit spatial component, exhibits greater bias and variability, highlighting its limitations in modelling spatial dependence.

Overall, these results indicate that the CAR model performs most reliably when spatial dependence is driven by areal structures, while the SAR model provides advantages under mixed dependence settings. The GLM yields stable but spatially naïve estimates, performing adequately only when spatial dependence is weak or absent.

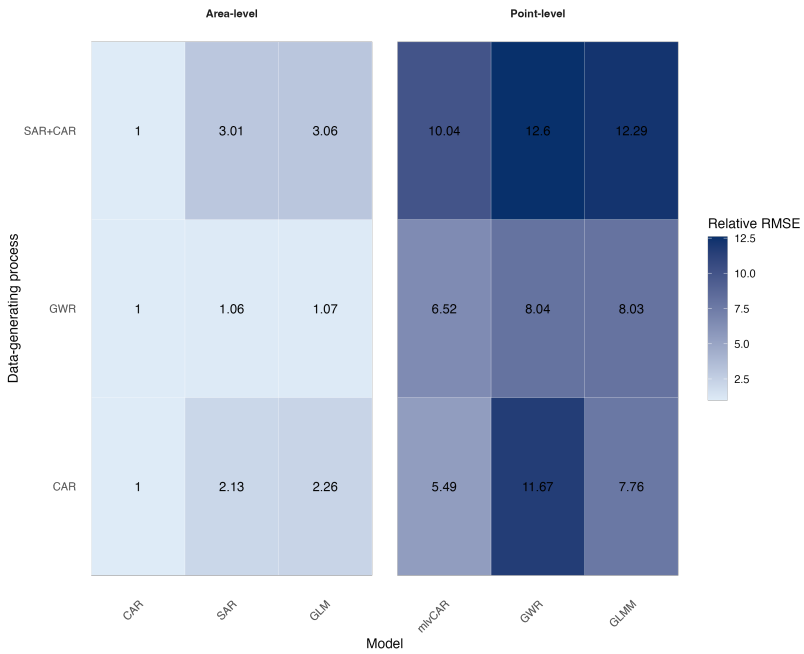

FIGURE 7. Heatmap of relative RMSE for competing models under different spatial data-generating processes. Results are shown for both areal and point-level data, with values normalised within each scenario (lower values indicate better performance).

Figure 7 presents relative RMSE heatmap for all models under three data-generating processes, evaluated separately for area-level and point-level datasets. Lower RMSE values indicate better predictive performance.

For the area-level data, the CAR model consistently exhibits the best performance across all data-generating processes, as indicated by the lowest relative RMSE in each scenario. This confirms its strong suitability when spatial dependence is primarily driven by areal random effects. Under the CAR DGP, the CAR model clearly outperforms both GLM and SAR, while under the SAR+CAR DGP it remains the most accurate, indicating robustness to more complex spatial structures. In contrast, under the GWR DGP, all models show comparable relative performance, suggesting only limited loss in accuracy when the true spatial process is continuous rather than discrete.

For the point-level data, the mlvCAR model demonstrates the strongest overall performance, achieving the lowest relative RMSE across all scenarios. Its advantage is evident under both the CAR and SAR+CAR DGPs, where it consistently outperforms GLMM and GWR. Although all models experience a degradation in performance under the GWR DGP, the mlvCAR model remains the most stable, indicating greater robustness to different forms of spatial dependence. Overall,

these results highlight the benefit of incorporating hierarchical spatial structure when modelling point-level data.

Overall, these findings suggest that:

- CAR-based models (CAR and mlvCAR) are most effective when spatial dependence arises from areal or hierarchical structures;
- the SAR model performs moderately well but does not outperform CAR;
- GWR and GLMM are less effective in the presence of discrete or mixed spatial dependence;
- CAR (area-level) and mlvCAR (point-level) emerge as the most reliable models across varying spatial data structures.

3.2. **Insights from ρ Values Across Data Scenarios.** Figure 8 illustrates the distribution of the estimated spatial autocorrelation parameter (ρ) across the three data-generating scenarios (CAR, SAR+CAR, and GWR). Each boxplot summarises results from 1000 simulation replications. Note that only three models—CAR, mlv-CAR, and SAR—include the spatial dependence parameter ρ.

The CAR model consistently produces high ρ estimates, with median values around 0.6–0.7, particularly under the CAR DGP. This confirms its effectiveness in capturing areal spatial dependence.

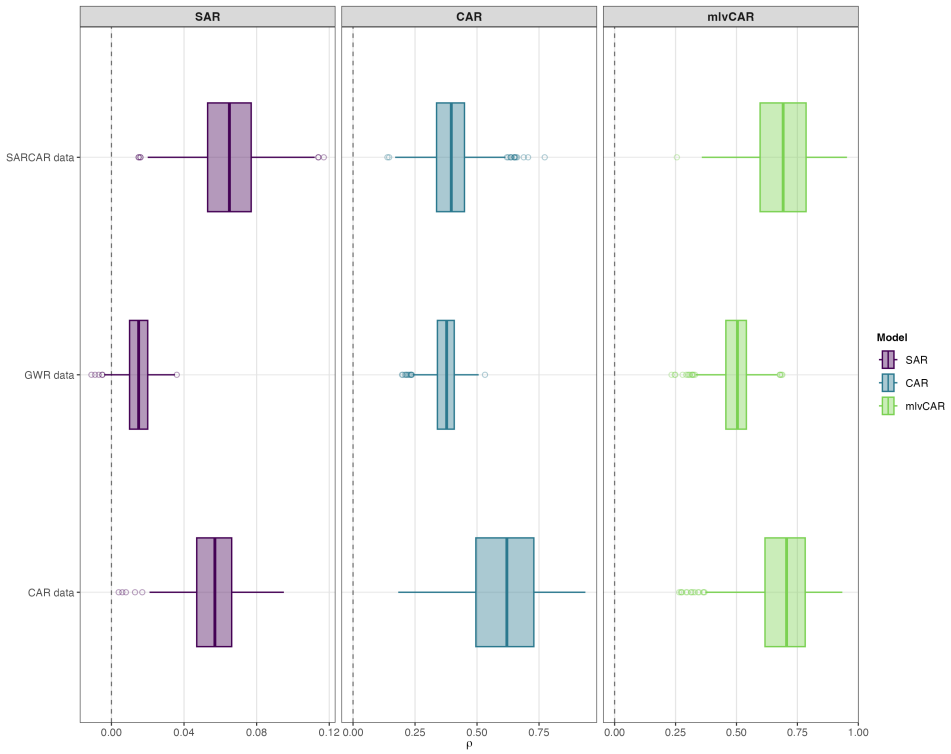

FIGURE 8. Distribution of ρ estimates across models and scenarios

The mlvCAR model exhibits a similar pattern but with a wider spread of ρ values, reflecting its hierarchical structure that captures both within- and between-area dependence. This additional flexibility allows it to accommodate more complex spatial relationships.

In contrast, the SAR model yields substantially lower ρ estimates, indicating that lag-based dependence is less suited for areal data where correlation arises from shared boundaries. Interestingly, even under the GWR DGP, both CAR and mlvCAR detect meaningful spatial dependence, suggesting that adjacency-based models can partially capture smooth spatial variation, albeit in an aggregated form.

The results highlight the importance of aligning the modelling framework with the underlying spatial structure of the data. Models that correctly reflect the data-generating process, such as CAR for areal data and mlvCAR for point-level data, consistently achieve the best performance. However, the findings also demonstrate that certain models exhibit greater robustness to misspecification. In particular, the mlvCAR model maintains relatively stable performance across all scenarios, suggesting that hierarchical spatial formulations provide a flexible framework when the true spatial mechanism is unknown.

Furthermore, the comparison between area-level and point-level analyses underscores the impact of data aggregation. The loss of spatial detail at the areal level can lead to different model behaviour and performance, reinforcing the importance of considering data resolution in spatial modelling. From a practical perspective, these results suggest that while simpler models such as CAR remain effective for areal data, more flexible hierarchical approaches are advantageous for point-referenced data and in situations where spatial structure is complex or uncertain.

Data Availability Statement. The data used to support the findings of this study are available from the corresponding author upon request.

Declarations. A conflict of interest, also known as a competing interest, is a situation in which an interest or connection -direct or indirect-could influence your research.

Author Contributions.

Indira Puteri Kinasih: Conceptualization, writing-original draft, visualization, software, validation. Haziq Jamil: Conceptualization, methodology, Data curation, formal analysis, resources, writing-review & editing, data curation. Elvynna Leong: writing-review & editing, methodology.

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